THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET A

Chapter 11: **Electricity** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The SI unit of electric current is: 1

(a) ohm **(b)** ampere **(c)** coulomb **(d)** volt

2. A charge of 10 C flows through a wire in 5 s. The current is: 1

(a) 15 A **(b)** 50 A **(c)** 2 A **(d)** 0.5 A

3. The resistance of a conductor of length l and area A is proportional to: 1

(a) $1/(lA)$ **(b)** lA **(c)** l/A **(d)** A/l

4. A student states: "Voltmeter is always connected in series." The correct response is: 1

(a) Correct **(b)** Incorrect — it is connected in parallel

(c) Correct for AC only **(d)** Cannot be determined

5. For three resistors of 2 Ω, 3 Ω and 5 Ω in series, Student A says the total is 10 Ω; Student B says 0.97 Ω. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

6. The heating effect of current is given by: 1

(a) $H = IRt^2$ **(b)** $H = IR^2t$ **(c)** $H = I^2Rt$ **(d)** $H = V/R$

7. 1 kWh is equal to: 1

(a) 3.6×10^3 J **(b)** 3.6×10^6 J **(c)** 1000 J **(d)** 3600 J

8. The resistance of a bulb rated 220 V, 100 W is: 1

(a) 22 000 Ω **(b)** 0.45 Ω **(c)** 2.2 Ω **(d)** 484 Ω

9. The material used for the filament of an electric bulb is: 1

(a) aluminium **(b)** tungsten **(c)** copper **(d)** nichrome

10. The potential difference across a 5 Ω resistor carrying 2 A is: 1

(a) 2.5 V **(b)** 7 V **(c)** 0.4 V **(d)** 10 V

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State Ohm's law. 2

12. Define 1 volt. 2

13. Why are alloys used in heating devices? 2

14. A student says that household appliances are connected in series. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Three resistors of 4 Ω, 6 Ω and 12 Ω are connected in parallel. Find the equivalent resistance and the current drawn from a 12 V battery. 3

16. An electric iron of 1 kW is used for 2 hours daily. Find the energy consumed in 30 days and the cost at ₹6 per unit. 3

17. Find the heat produced in 10 s by a current of 2 A through a resistor of 50 Ω. 3

OR

What is resistivity? On what factors does the resistance of a conductor depend? 3

18. Draw (describe) a circuit diagram with a cell, a resistor, an ammeter, a voltmeter and a plug key. How are the ammeter and voltmeter connected? 3

OR

A wire of resistance 10 Ω is stretched to double its length. Find its new resistance (volume constant). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Derive the expression for equivalent resistance of three resistors in series.
(ii) Two resistors of 5 Ω and 10 Ω are connected in series to a 6 V battery. Find the current and the potential difference across each. 5

OR

(i) Derive the expression for equivalent resistance of resistors in parallel. 5
(ii) Why is a parallel arrangement better for household circuits?

20. **Case study:** A house has 4 bulbs of 60 W each used 5 h daily and a 1000 W heater used 2 h daily at 220 V. 5
(i) Find the total energy consumed per day in kWh.
(ii) Find the monthly (30 days) cost at ₹5 per unit.
(iii) Find the current drawn by the heater.
(iv) Why is a fuse used?

OR

(i) Explain Joule's law of heating. 5
(ii) State two applications of heating effect.
(iii) An electric heater draws 5 A from a 220 V line. Find its power and energy used in 2 h.

END OF QUESTION PAPER — Total: 40 Marks

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